Homework-5

A single degree-of-freedom (SDOF) structure is represented with a truss element connected to a lumped mass of $M = 360$ ton. The truss material is modeled using Menegotto-Pinto material model with the following parameters:

$G_y = 0.2$ MPa

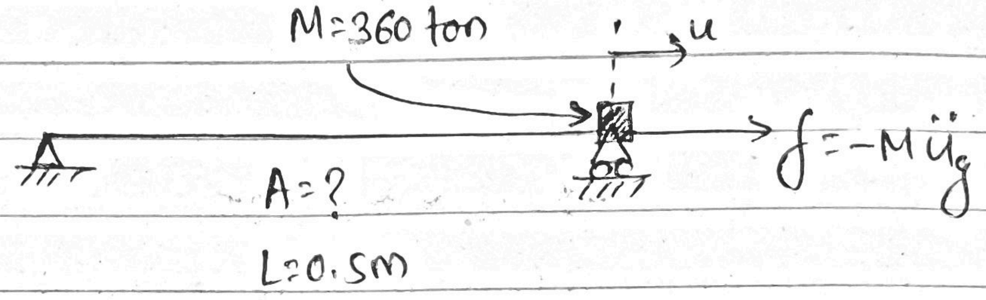
$E = 200$ GPa

$b = 0.05$

$R_0 = 20$

$R_1 = 0.925$

$a_1 = a_2 = 0$



Consider the included Northridge seismic record as the ground motion ~~($\ddot{u}_g$)~~ ($\ddot{u}_g$).

(1) Determine the area of the truss element to have an initial (linear-elastic) period of $T_0 = 0.15$ s.

→ Soln.

we have,

$$T_0 = 2\pi\sqrt{\frac{M}{K_0}}$$

$$\Rightarrow 0.15 = 2\pi\sqrt{\frac{360 \times 10^3}{K_0}}$$

$$\Rightarrow K_0 = 631.65 \times 10^6 \text{ N/m}$$

$$= 631.65 \text{ MN/m}$$

Also,

$$K_0 = \frac{AE}{L}$$

$$\Rightarrow 631.65 \times 10^6 = \frac{A \times 200 \times 10^9}{0.5}$$

$$\Rightarrow A = 1.58 \times 10^{-3}\ m^2$$

$$\approx 1580\ mm^2$$

(2) Calculate the damping coefficient considering a constant damping ratio of 2%.

→ Soln.

we have,

$$c = 2 \xi M \omega_0$$

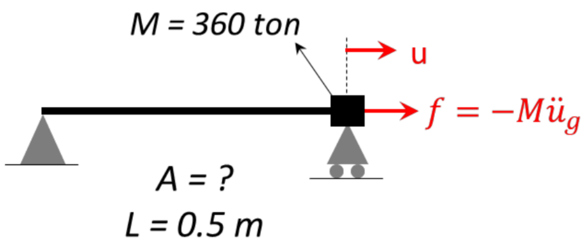
$$= 2 \times \frac{2}{100} \times 360 \times 10^3 \times 2\pi \times \frac{1}{0.15}$$

$$= 603.186 \times 10^3\ kg/s$$

$$= 603.19\ kN\text{-}s/m$$

Code for Problem 3, 4, and 5: https://github.com/nirajkyadav/Homeworks/blob/main/HW5/Q3.py

Problem # 3



3. Implicit Nonlinear Dynamic Analysis: Write a Matlab code to perform time integration using Newmark-beta method with Newton-Raphson iterations – select a suitable convergence criterion based on your experience. Follow these steps:
   a. Perform the nonlinear time history analysis using both constant average and linear acceleration methos with a time step size of $\Delta t = 0.02\ sec$. Plot the displacement and acceleration responses of the structure and compare the results between the two methods. State your observations.
   b. Resample the input time history using the "resample" command in Matlab with $\Delta t = 0.01\ sec$, $\Delta t = 0.005\ sec$, and smaller step sizes if needed (first read about the command from Matlab Help to understand the details and make sure you are using it correctly). Perform a time-step convergence analysis using the constant average acceleration method. Plot your results and suggest a suitable time-step size for the constant average acceleration method in this problem.

Area = 1579.137 mm^2, Damping Coefficient = 603.186 kN.s/m

Reference response (converged):

peak |u| ≈ 85.6 mm, peak |a| ≈ 8.24 m/s^2. Both implicit ($\Delta t \to 0$) and explicit ($\Delta t \to 0$) agree to < 0.04% , which validates the two independent integrators.

Problem 3a, Implicit Nonlinear Dynamic Analysis, dt = 0.02 s

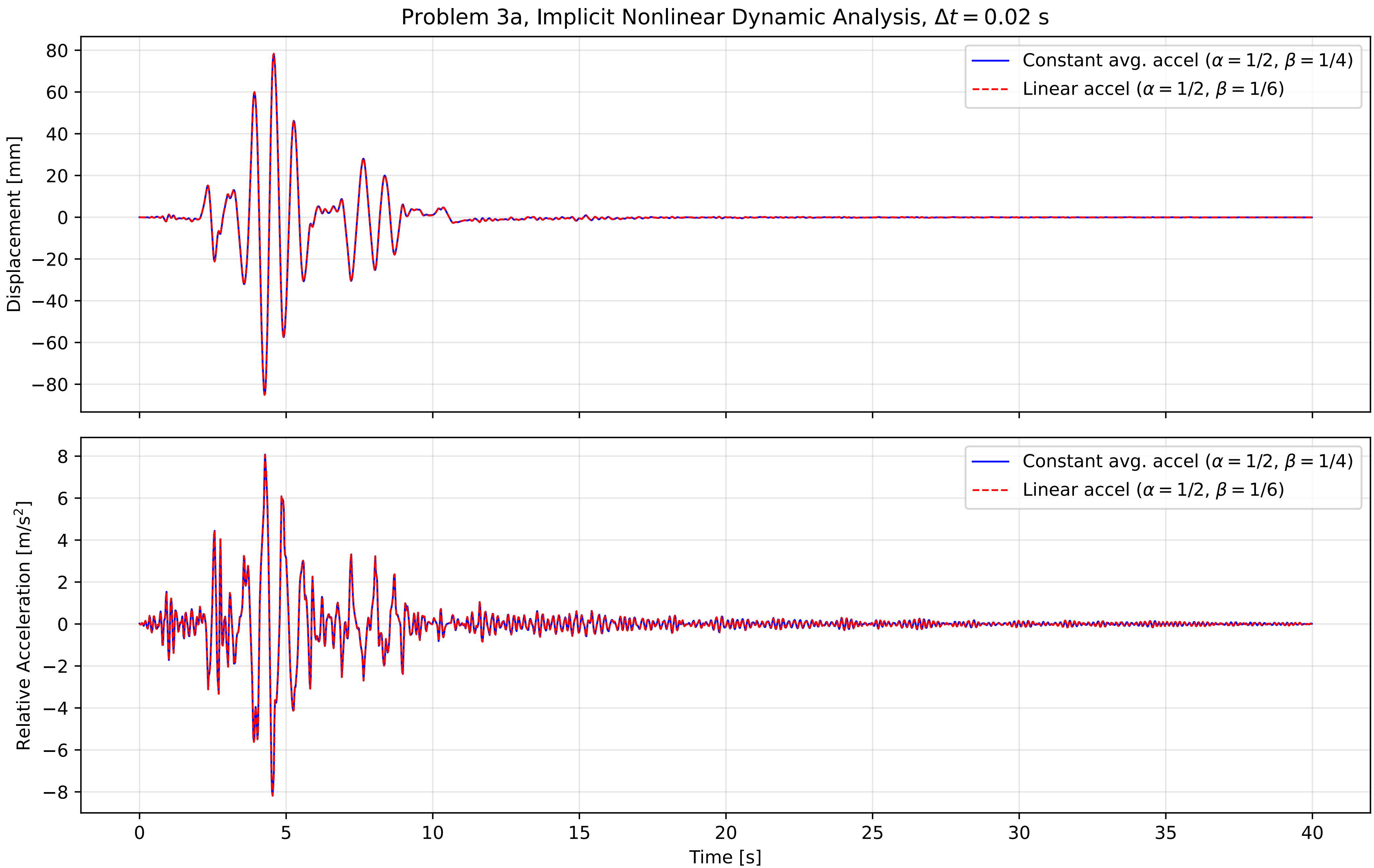
Constant Average Acceleration Method : peak|u| = 84.845 mm, peak|a| = 8.150 m/s^2
Linear Acceleration Method : peak|u| = 85.111 mm, peak|a| = 8.188 m/s^2

Both analyses run stably. Linear acceleration method (conditionally stable) is more accurate than constant average acceleration method (unconditionally stable) when compared to reference response, but the difference is almost negligible at this time step.

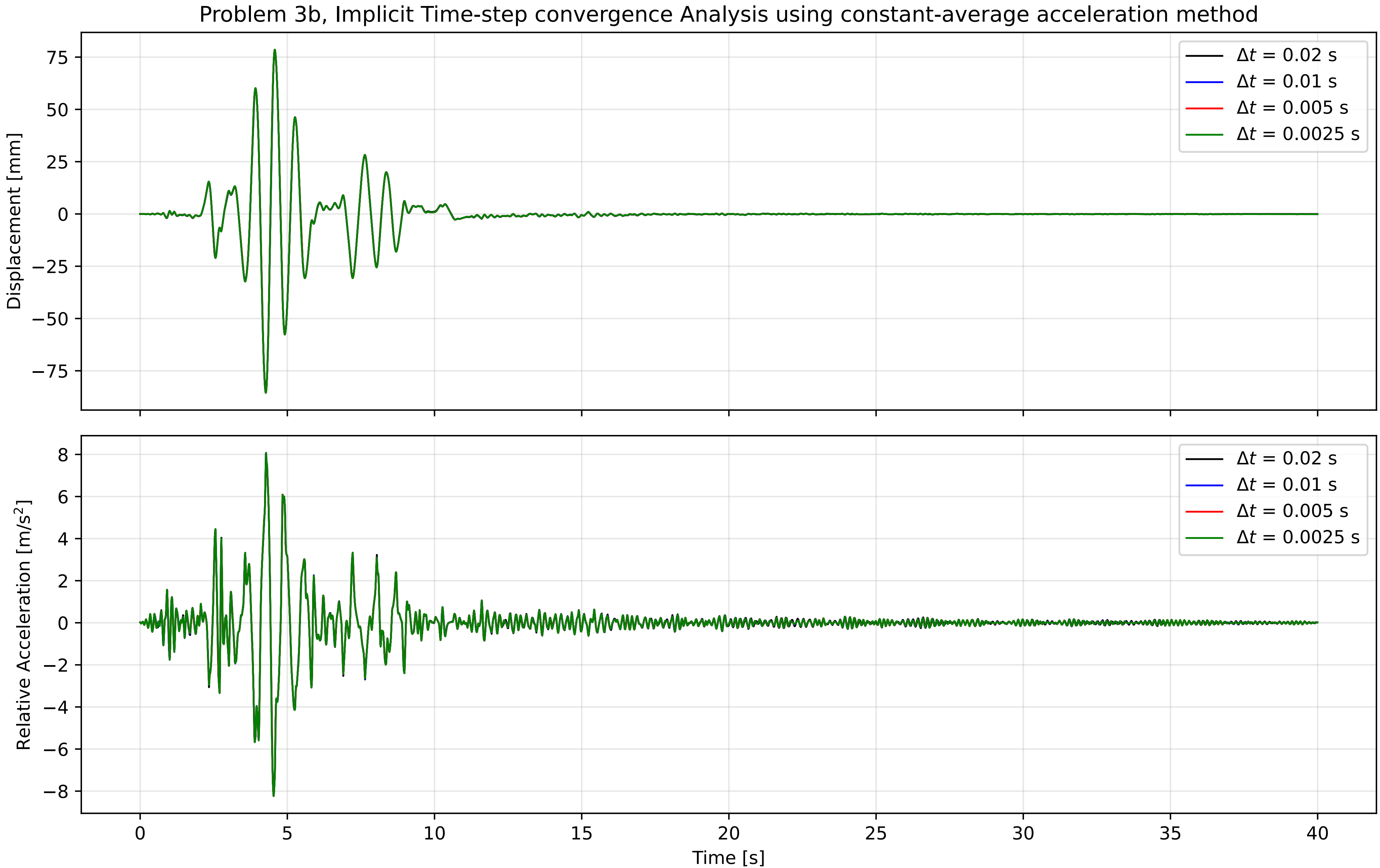
Problem 3b, Implicit Time-step convergence Analysis,

constant-average acceleration method, resampling performed using MATLAB

dt (s)	**Peak|u| (mm)**	**Peak|a| (m/s^2)**
0.02	84.845	8.150
0.01	85.417	8.210
0.005	85.517	8.238
0.0025	85.568	8.239



Problem 3a, Implicit Nonlinear Dynamic Analysis, $\Delta t = 0.02$ s
Displacement [mm]
Constant avg. accel ($\alpha = 1/2$, $\beta = 1/4$)
Linear accel ($\alpha = 1/2$, $\beta = 1/6$)
Relative Acceleration [m/s^2]
Constant avg. accel ($\alpha = 1/2$, $\beta = 1/4$)
Linear accel ($\alpha = 1/2$, $\beta = 1/6$)
Time [s]

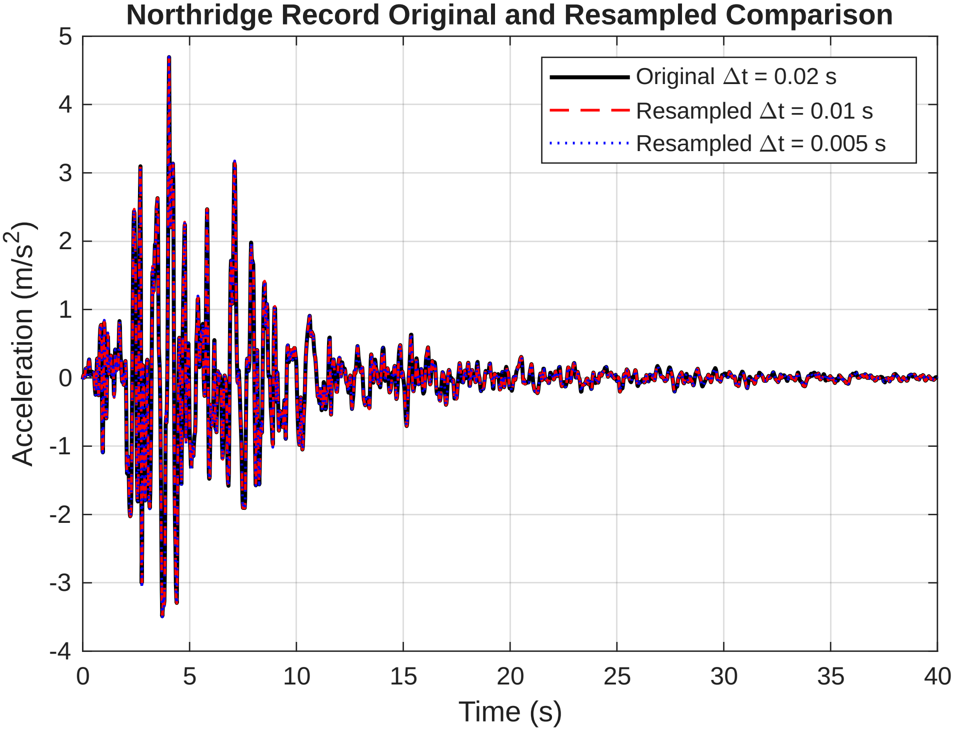

Problem 3b, Implicit Time-step convergence Analysis using constant-average acceleration method
Displacement [mm]
Relative Acceleration [m/s^2]
Time [s]
Δt = 0.02 s
Δt = 0.01 s
Δt = 0.005 s
Δt = 0.0025 s
75
50
25
0
−25
−50
−75
8
6
4
2
−2
−4
−6
−8
5
10
15
20
25
30
35
40

All the analyses run stably as constant average acceleration method is unconditionally stable. Considering the number of steps, dt = 0.005s have an error of 0.1% both for displacement and acceleration. Thus, a time step of 0.005s is suggested.

Matlab Code used for Resampling:

Method: y = resample(x,p,q) resamples the input sequence, x, at p/q times the original sample rate. resample applies an FIR Antialiasing Lowpass Filter to x and compensates for the delay introduced by the filter. The function operates along the first array dimension with size greater than 1 [https://www.mathworks.com/help/signal/ref/resample.html]

```
% Importing the data
filename = 'Northridge Record.txt';
data = readtable(filename);

% Time and acceleration
t = data{:, 1};
acc = data{:, 2};

dt_original = 0.02; % Original time step = 0.02 sec
dt_target1 = 0.01;  % Target time step   = 0.01 sec
dt_target2 = 0.005; % Target time step   = 0.005 sec

% Ratio = f_target / f_original = (1/dt_target) / (1/dt_original) = dt_original / dt_target

% For dt = 0.01
[p1, q1] = rat(dt_original / dt_target1);  % Resampling ratio
acc1 = resample(acc, p1, q1);
t1 = (0:length(acc1)-1)' * dt_target1;     % Time column vector for 0.01
output1 = table(t1, acc1, 'VariableNames', {'time_sec', 'Acc_m_s2'});
writetable(output1, 'Northridge_0_01.txt', 'Delimiter', '\t');

% For dt = 0.005
[p2, q2] = rat(dt_original / dt_target2);  % Resampling ratio
acc2 = resample(acc, p2, q2);
t2 = (0:length(acc2)-1)' * dt_target2;     % Time column vector for 0.005
output2 = table(t2, acc2, 'VariableNames', {'time_sec', 'Acc_m_s2'});
writetable(output2, 'Northridge_0_005.txt', 'Delimiter', '\t');

% Comparison Plots
figure;
plot(t, acc, 'k-', 'LineWidth', 1.5); hold on;
plot(t1, acc1, 'r--', 'LineWidth', 1);
plot(t2, acc2, 'b:', 'LineWidth', 1);
grid on;
xlabel('Time (s)');
ylabel('Acceleration (m/s^2)');
legend('Original \Deltat = 0.02 s', ...
  'Resampled \Deltat = 0.01 s', ...
  'Resampled \Deltat = 0.005 s');
title('Northridge Record Original and Resampled Comparison');
```



Northridge Record Original and Resampled Comparison
Original Δt = 0.02 s
Resampled Δt = 0.01 s
Resampled Δt = 0.005 s
Acceleration (m/s²)
Time (s)
5
4
3
2
1
0
-1
-2
-3
-4
0
5
10
15
20
25
30
35
40

Problem # 4

4. Explicit Nonlinear Dynamic Analysis: Write a Matlab code to perform explicit analysis. Follow these steps:
   a. Select an initial time-step size based on the CFL condition discussed in the class and perform the analysis.
   c. Perform a time-step convergence analysis to understand the effect of time-step size. For this purpose, increase and decrease the time-step size derived from the CFL condition and state your observations. Suggest a suitable time-step size for this problem.

Using Lumped Mass Period,

$$\Delta t \leq \Delta t_{critical} = \frac{Tmin}{\pi} \left(\sqrt{1 - \xi^{2}} - \xi\right)$$

$\Delta t_{LM} \leq (0.15/\pi)\ ((1\text{-}0.02^2)^{0.5} - 0.02) = 0.04678$s

Using CFL condition,

Density of Steel (ϱ) = 7850.0 kg/m^3

Mass of the bar = $7860\times0.5\times1580\times10^{-6} \approx 6.2$ kg

Velocity of wave (c) = $(E / \varrho)^{0.5} = (200\times10^9/7850)^{0.5} = 5047.54$ m/s

$\Delta_{\text{critical CFL}}$ = L/c = 0.5/5047.54 = 9.9×10^{-5} s $\approx 10^{-4}$ s

Considering CFL condition, an initial time step of 0.00005s is adopted for Problem 4a. The bar carries negligible mass in the equations of motion (6.2 kg vs 360,000 kg of lumped mass).

Problem 4a, Explicit Nonlinear Dynamic Analysis

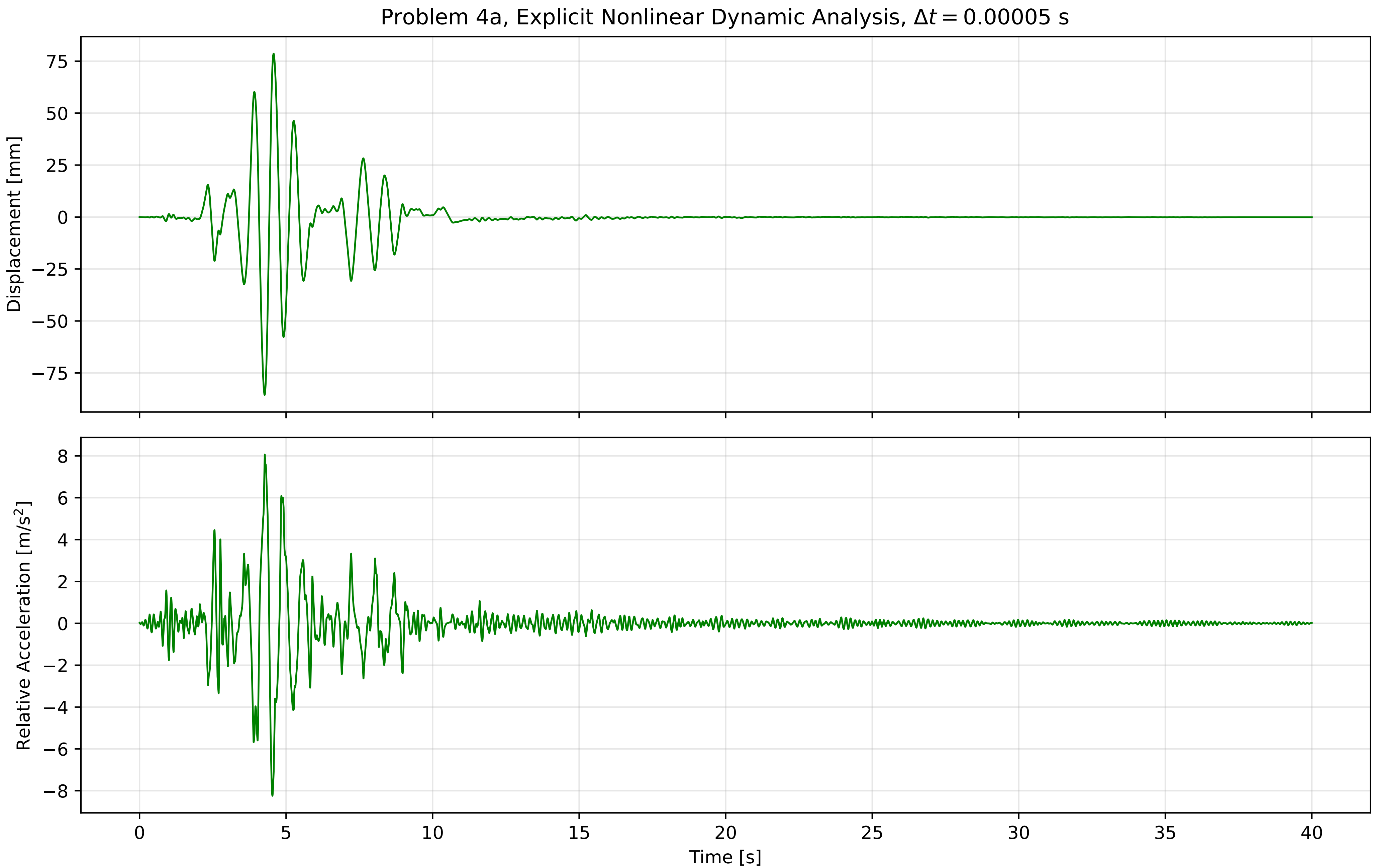
Critical Time Step based on CFL condition = 0.000099 s

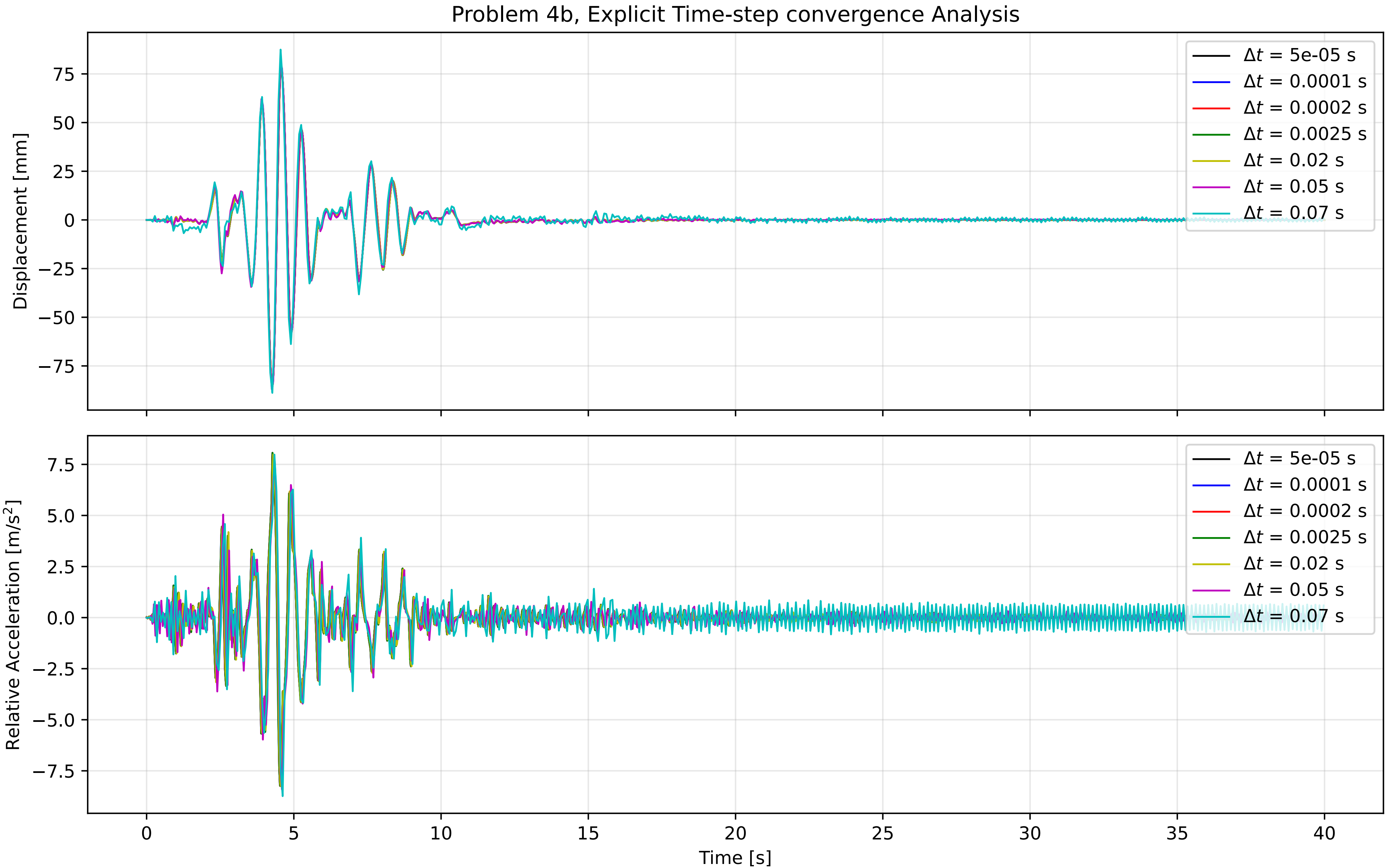
Explicit Analysis, peak|u| = 85.575 mm, peak|a| = 8.242 m/s^2

Problem 4b, Explicit Time-step convergence Analysis

dt (s)	Peak|u| (mm)	Peak|a| (m/s^2)
0.00005	85.575	8.242
0.0001	85.575	8.242
0.0002	85.575	8.242
0.0025	85.579	8.241
0.02	85.608	8.261
0.05	86.930	8.422
0.07	88.855	8.740

As observed from the results, even when $\Delta t \geq \Delta_{\text{critical CFL}}$, results for peak displacement and acceleration are identical. This is because, in this system of lumped-mass SDOF, the bar carries no mass in the equations of motion (its self-mass is ≈ 6 kg vs 360,000 kg). As observed, the results are stable up to Δt_{LM} and diverges above it (i.e. 0.05s and 0.07s). At dt = 0.0025s both the errors are ≤ 0.1% for displacement and acceleration. Thus, a time step of 0.0025s is suggested.


Problem 4a, Explicit Nonlinear Dynamic Analysis, Δt = 0.00005 s
Displacement [mm]
75
50
25
0
−25
−50
−75
Relative Acceleration [m/s²]
8
6
4
2
0
−2
−4
−6
−8
0
5
10
15
20
25
30
35
40
Time [s]


Problem 4b, Explicit Time-step convergence Analysis
Displacement [mm]
Relative Acceleration [m/s^2]
Time [s]
Δt = 5e-05 s
Δt = 0.0001 s
Δt = 0.0002 s
Δt = 0.0025 s
Δt = 0.02 s
Δt = 0.05 s
Δt = 0.07 s

Problem # 5

5. Compare the displacement and acceleration responses of the structure from the most-efficient implicit and explicit methods. Compare the total computation time between the two methods. State your observations.

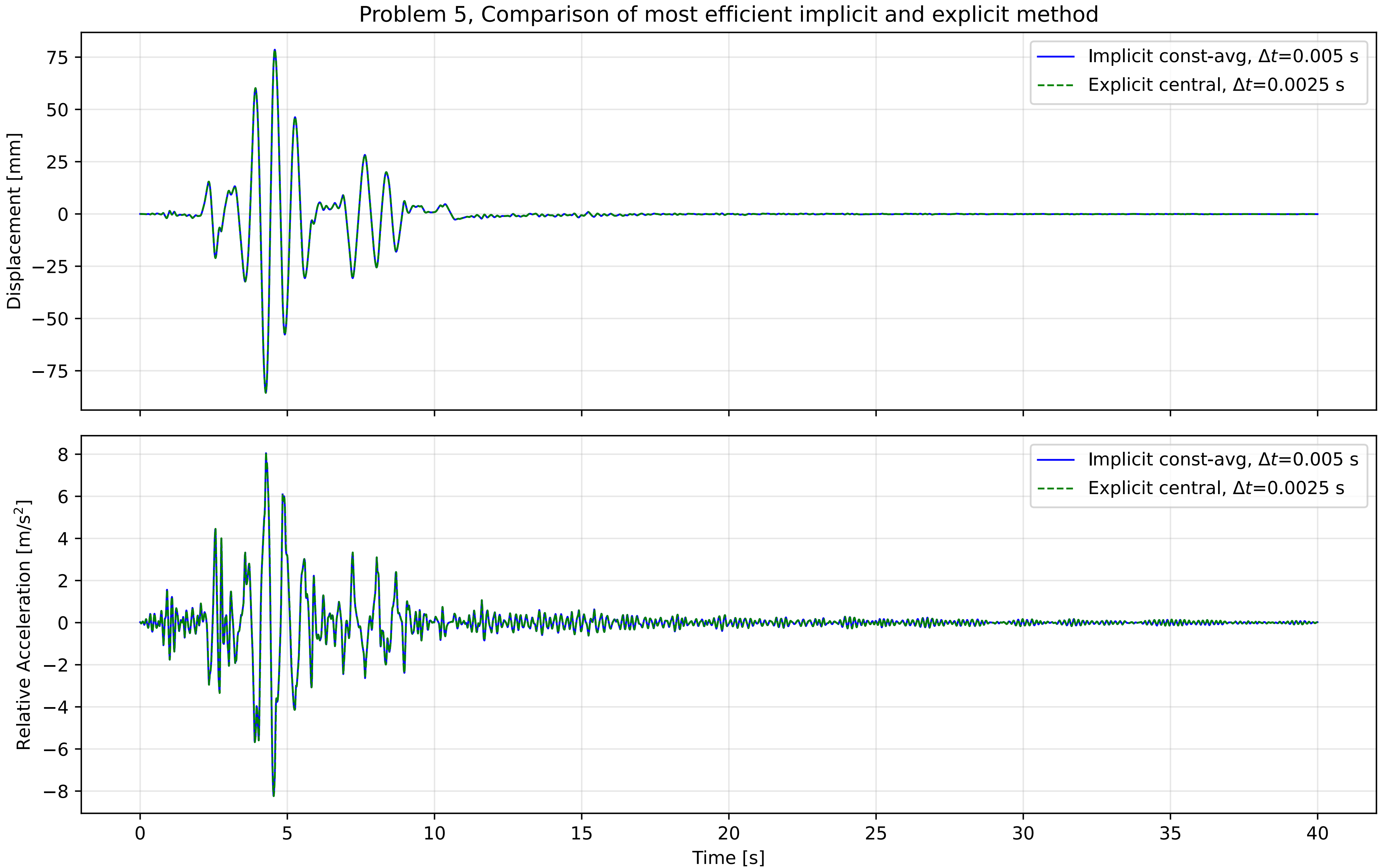
Implicit (const-avg, dt=0.005):

peak |u|= 85.517 mm, peak |a|= 8.238 m/s^2, steps = 8000, Total computation time = 42.0 ms

Explicit (central,   dt=0.0025):

peak |u|= 85.579 mm, peak |a|= 8.241 m/s^2, steps = 16000, Total computation time =70.2 ms

Both methods give very close results. Per step, explicit is faster as no iteration is required but requires far more total steps while implicit requires iteration with fewer total steps. For this single-DOF problem the two are within a small factor when each uses its proper step.


Problem 5, Comparison of most efficient implicit and explicit method
Displacement [mm]
Implicit const-avg, Δt=0.005 s
Explicit central, Δt=0.0025 s
Relative Acceleration [m/s²]
Implicit const-avg, Δt=0.005 s
Explicit central, Δt=0.0025 s
Time [s]